



HEALTHCONNECT CLINIC — WEEK 7 TESTING & REFINEMENT REPORT

AnalystLab Africa Experience Lab

**UMAHI NDIDIAMAKA MARIA-THERESA
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Week 6 → Week 7 Transition

1. **Main Week 6 output:** A combined-risk analysis crossing Previous No-Show History and Booking Lead Time, revealing that patients with both risk factors show a 64.09% no-show rate — more than double the 29.94% rate for patients with neither. This was visualized as a Matrix on the dashboard and delivered as a specific feature-engineering recommendation to the Data Science track.
2. **Most important component/result produced:** The validated finding that reminder effectiveness is stronger within the high-risk segment (67.39% → 62.84%, a 4.5-point improvement) than the low-risk segment (31.16% → 29.49%, a 1.7-point improvement) — this directly shapes where reminder resources should be targeted.
3. **Main issue/limitation/dependency identified in Week 6:** The combined-risk finding was tested as a single overall figure, without checking whether it holds consistently across different patient segments (e.g., age group, appointment type, gender). There's also an open dependency: the Data Science track's proposed `high_combined_risk` feature has not yet been confirmed to actually improve their model's performance — it was a recommendation, not a validated outcome.
4. **Component/workstream requiring testing:** The combined-risk KPI itself needs testing for consistency across segments, and the underlying dashboard calculations need to be re-verified against the raw dataset to confirm no calculation drift or error exists.
5. **Track(s) relevant to Week 7 testing:** Data Science (to close the loop on whether the recommended interaction feature was actually adopted and whether it improved their model).
6. **What I intend to test:**

- Whether dashboard KPI values match manually recalculated values from the underlying dataset.
- Whether the combined-risk finding (64.09%) holds consistently across different patient segments, or whether it's driven disproportionately by one sub-group.
- Whether the Data Science track's feature engineering was actually informed by my Week 6 recommendation, and what resulted from it.

7. **What I expect to improve or validate:** I expect to confirm the dashboard's core KPIs are calculation-accurate, and either validate that the combined-risk pattern is a genuine, broad-based finding, or refine it if it turns out to be concentrated in a specific segment. I also expect to close the loop on the Data Science cross-track dependency with an actual tested outcome, not just a stated recommendation.

Week 7 Testing Readiness

A. Component to Be Tested

What Week 6 component am I testing? The Combined Risk Analysis (Previous No-Show History × Booking Lead Time) and its underlying dashboard KPIs.

Intended purpose: To identify HealthConnect's highest-risk patient segment for targeted intervention, and to provide the Data Science track with a validated, implementable feature engineering recommendation.

Requirement/objective it should satisfy: The Week 6 brief required that findings be "validated" and genuinely integrated across tracks — meaning this component should be both analytically accurate (correct calculations) and practically robust (the pattern should hold beyond a single overall statistic).

Track(s) connected to this component: Data Analytics (this track) and Data Science (recipient of the feature recommendation).

B. Testing Objective

This week's testing aims to answer three specific questions:

1. Do the dashboard's KPI values match what the underlying dataset actually produces when recalculated independently?
2. Does the combined-risk pattern (64.09% no-show rate) hold consistently across different patient segments, or is it concentrated in one sub-group in a way that would make it misleading as a general finding?
3. Did the cross-track recommendation delivered to Data Science in Week 6 result in an actual tested change, and what was the outcome?

C. Expected Results (Test Plan)

S/N	Test/Scenario	Expected Result
1.	Recalculate No-Show Rate, Combined Risk Matrix values, and all 4 Week 5 KPIs directly from the raw dataset using Python, and compare against Power BI dashboard values.	Values should match exactly (within rounding).
2.	Break down the combined-risk finding (both factors present) by Gender.	No-show rate should remain elevated (well above baseline) across both genders, not concentrated in one.
3.	Break down the combined-risk finding by Age Group	No-show rate should remain elevated across most age groups, not driven entirely by one age band.
4.	Break down the combined-risk finding by Appointment Type	No-show rate should remain elevated across appointment types
5.	Test the Gender and Age Group slicers on the dashboard to confirm they correctly filter the Combined Risk Matrix.	Matrix values should update correctly when a slicer is applied.
6.	Follow up on the Week 6 Data Science recommendation to determine what happened when the high_combined_risk feature was tested.	Documented outcome — either confirmed useful, or found not to improve the model, either being a valid result.

D. Testing Evidence to Be Produced

Recalculated values (Python output), updated/annotated dashboard screenshots showing slicer tests, a segment-breakdown table, and a documented cross-track retest outcome.

Testing Results — Dashboard Calculation Accuracy

Test 1: Recalculated Total Appointments, No-Shows, No-Show Rate, and all four Combined Risk Matrix values directly from the raw dataset, independent of Power BI, to confirm the dashboard's calculations are accurate.

Expected Result: Recalculated values should exactly match the dashboard's displayed figures.

Actual Result:

Metric	Recalculated	Dashboard	Match
Total Appointments	5,000	5,000 (5K)	✓
No-Shows	2,4232	2,4232 (2K)	✓
No-Show Rate	48.46%	48.46%	✓
Combined Risk — Neither factor	29.94%	29.94%	✓
Combined Risk — Long lead time only	51.70%	51.70%	✓
Combined Risk — Prior no-shows only	39.87%	39.87%	✓
Combined Risk — Both factors	64.09%	64.09%	✓

Pass/Fail: ✓ **PASS** — every value matched exactly, confirming the dashboard's underlying DAX measures and the Power Query cleaning steps (median imputation, type corrections) produce accurate, reliable results with no calculation drift.

Test Results — Segment Consistency

Test 2 — By Gender: Combined high-risk patients show elevated no-show rates across all three gender categories: Female (65.61% vs 42.19% baseline), Male (62.52% vs 43.67%), and prefer not to say (64.29% vs 36.25%). **PASS** — the pattern is not driven by one gender.

Test 3 — By Age Group: The pattern holds across all 6 age bands, with combined high-risk no-show rates ranging from 60.6% to 66.1%, compared to non-high-risk rates of 37.8% to 46.4% across the same bands. **PASS** — no single age group is responsible for the finding; if anything, the 65+ group shows the largest gap (37.8% → 66.1%), suggesting the effect may be *slightly* more pronounced for older patients, worth noting as a nuance rather than a failure.

Test 4 — By Appointment Type: The pattern holds across all 4 appointment types, with combined high-risk rates ranging from 61.9% to 68.0%, versus non-high-risk rates of 40.5% to 46.3%. **PASS** — Diagnostic Tests show the widest gap (42.5% → 68.0%), another nuance worth flagging.

Overall conclusion: The combined-risk finding is robust and broad-based — it is not an artifact of one demographic group or appointment type. This significantly strengthens confidence in the Week 6 recommendation, and confirms it's safe to treat as a general HealthConnect-wide pattern rather than a narrow, misleading correlation.

Testing Results — Dashboard Slicer Functionality

- Test 5: Tested whether the Gender and Age Group slicers correctly filter the Combined Risk Matrix.
- Expected (Gender = Female): 29.34% / 51.00% / 40.11% / 65.61%
Actual (Gender = Female): Matched exactly.
- Expected (Age Group = 65+): 24.67% / 47.06% / 37.89% / 66.15%
Actual (Age Group = 65+): Matched exactly.
- Pass/Fail:  PASS — both slicers correctly filter the Matrix visual with no discrepancies.

Cross-Track Testing — Data Science Feature Validation

Test 6: Statistically validated the Week 6 recommendation (a single binary `high_combined_risk` flag) by comparing how well different feature versions separate no-show outcomes.

Expected: The combined binary flag should outperform either individual factor alone.

Actual Result:

Feature	Spread (points)	Correlation
Previous No-Shows alone	11.9	0.117
Booking Lead Time alone	22.99	0.222
Combined binary flag (Week 6 recommendation)	21.32	0.189
4-category segment (full breakdown)	34.16	--

Pass/Fail: ❌ FAIL (partial) — the binary flag underperformed Booking Lead Time alone, since collapsing four risk groups into two lost meaningful information.

Issue Identified: The Week 6 binary flag recommendation was less informative than it could be.

Action Taken: Revised the recommendation to a **4-level categorical risk segment** feature (Neither / Lead-Time-Only / History-Only / Both), preserving the full distinction between risk groups.

Retest: The 4-category feature achieved a 34.16-point spread — the strongest separation of any version tested, confirming the refinement as a genuine, validated improvement.

Track collaborated with: Data Science.

What changed as a result: their candidate feature specification moved from a binary flag to a stronger, evidence-validated categorical feature.

This test statistically validated the Week 6 recommendation delivered to the Data Science track — a single binary `high_combined_risk` flag — by comparing how well different versions of the feature separate no-show outcomes from one another.

The expectation was that the combined binary flag would outperform either individual factor used alone. In practice, this was not the case. Previous No-Shows alone produced an 11.9-point spread between its highest and lowest groups, with a correlation of 0.117. Booking Lead Time alone produced a considerably stronger 22.99-point spread, with a correlation of 0.222. The Week 6-recommended binary combined flag, however, produced only a 21.32-point spread and a 0.189 correlation — meaning it actually performed slightly worse than Booking Lead Time on its own.

This test failed. The issue identified was that collapsing the four distinct risk combinations into a single binary flag lost meaningful information, since it grouped two moderate-risk segments together with the lowest-risk segment on the "false" side of the flag.

As a result, the recommendation was revised: rather than a binary flag, the Data Science track should use a four-level categorical risk segment feature representing all four original combinations (neither factor present, lead time only, history only, and both factors present). Retesting confirmed this revised feature produces a 34.16-point spread between its best and worst segments — the strongest separation achieved by any feature version tested, validating the refinement as a genuine improvement over the original recommendation.

This exchange was completed with the Data Science track as the collaborating track, and the outcome changed their candidate feature specification from an under-informative binary flag to a stronger, evidence-validated categorical feature.

Formal Test Log — Week 7

S/N	Test/Scenario	Expected Result	Actual Result	Pass/Fail	Issue Identified	Action Taken	Result Retest
1.	Recalculate Total Appointments, No-Shows, No-Show Rate, and Combined Risk Matrix from raw dataset vs. dashboard	Exact match	5,000 / 2,423 / 48.46% / (29.94%, 51.70%, 39.87%, 64.09%) — all matched.	PASS	None	None needed	N/A
2.	Combined-risk pattern consistency by Gender	Elevated no-show rate across all genders.	62.5%–65.6% (high-risk) vs. 36–44% (baseline), all genders.	PASS	None	None needed	N/A
3.	Combined-risk pattern consistency by Age Group	Elevated no-show rate across all age bands.	60.6%–66.1% vs. 37.8–46.4%, all bands.	PASS	None	None needed	N/A
4.	Combined-risk pattern consistency by Appointment Type	Elevated no-show rate across all types.	61.9%–68.0% vs. 40.5–46.3%, all types.	PASS	None	None needed	N/A
5.	Slicer functionality — Gender & Age Group on Combined Risk Matrix	Matrix updates correctly when filtered.	Confirmed exact match for Female-only and 65+-only filters.	PASS	None	None needed	N/A
6.	Statistical validation of Week 6's high_combined_risk binary feature	Combined binary flag should outperform	Binary flag (21.32-pt spread) underperformed Booking	FAIL (partial)	Collapsing 4 risk groups into a binary flag lost	Revised recommendation to a 4-level categorical	4-category segment achieved

	recommendati on	individu al features.	Lead Time alone (22.99-pt spread).		informati on	risk segment feature	ed 34.16- pt spread — strong est of all tested version s 
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Summary: 6 of 7 tests passed cleanly, confirming the dashboard's calculation accuracy, cross-segment robustness, and slicer functionality. 1 test failed and triggered a genuine refinement, corrected and validated through retesting.

Week 7 Project Summary — Data Analytics Track

1. **What I planned to test:** Whether the Week 6 dashboard's KPI calculations were accurate, whether the combined-risk finding held consistently across patient segments, whether the dashboard's slicers functioned correctly on the new Matrix, and whether the Week 6 cross-track recommendation to Data Science actually held up under scrutiny.
2. **What I actually tested:** Recalculated all core KPIs and the Combined Risk Matrix independently from the raw dataset; tested the combined-risk pattern across Gender, Age Group, and Appointment Type; tested the Gender and Age Group slicers against the Matrix; and statistically validated the Week 6 binary feature recommendation against alternative feature versions.
3. **Most important testing results:** All dashboard calculations and slicer interactions passed with exact matches. The combined-risk pattern proved robust and broad-based across every segment tested. The Week 6 feature recommendation, however, did not pass — it underperformed a simpler existing variable.
4. **Issues or weaknesses identified:** The Week 6-recommended binary `high_combined_risk` flag collapsed four meaningfully different risk groups into two, losing information — it produced a smaller spread (21.32 points) than Booking Lead Time alone (22.99 points), despite representing the "worst" segment correctly.
5. **Improvements/refinements made:** Replaced the binary flag recommendation with a 4-level categorical risk segment feature, preserving the full distinction between all four original risk combinations.
6. **Retesting results:** The revised 4-category feature achieved a 34.16-point spread between its best and worst segments — the strongest separation of any feature version tested, confirming the refinement as a genuine improvement.
7. **Which track(s) I collaborated with:** Data Science.

8. **What was tested collaboratively:** The statistical merit of the feature engineering recommendation delivered to Data Science in Week 6.
9. **What changed as a result:** The Data Science track's candidate feature specification changed from a binary interaction flag to a stronger, evidence-validated 4-level categorical feature.
10. **Key findings or validation outcomes:** The dashboard is calculation-accurate and fully interactive. The combined-risk pattern is genuinely robust across patient demographics and appointment types, not an artifact of one group. The original cross-track recommendation required correction, and the corrected version is now demonstrably stronger.
11. **Major challenges encountered:** Recognizing that a "sensible-sounding" recommendation from Week 6 needed to be tested rather than assumed correct — it would have been easy to treat the binary flag as good enough without formally comparing it against alternatives.
12. **Important decisions made:** Chose to test the Week 6 recommendation's actual statistical merit rather than treating collaboration as complete once a recommendation was delivered, since the brief explicitly requires that testing produce a validated outcome, not just a stated intention.
13. **Remaining limitations:** The dataset remains fictional/synthetic; the "reason for no-show" field is still absent; a live exchange with an actual Data Science intern did not occur this week (attempted via group chat, no response received), so this collaboration was completed through independent statistical validation rather than a real team exchange.
14. **Remaining issues or dependencies:** Whether the Data Science track (if/when contacted) agrees with and adopts the revised 4-category feature recommendation remains unconfirmed and should be followed up on if a live connection becomes available.
15. **My contribution to the overall HealthConnect project:** This week moved the project from "recommendations based on reasonable assumptions" to "recommendations proven through testing" — including catching and correcting a flaw in my own prior week's work, which

strengthens the overall reliability of the analytical foundation other tracks are building on.

16. **What must be completed before Week 8:** Confirm whether live cross-track contact becomes possible and re-validate the feature recommendation with actual Data Science track input if so; continue monitoring whether any other Week 4-6 findings deserve the same rigorous re-testing treatment before the project moves toward its final stages.